

Fuel Efficiency Analysis using Machine Learning Approach

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ABSTRACT:

In this modern era, vehicular improvements in terms of cost, effective utilization of fuel saving, power has become a necessary and prime task. The applications of predicting fuel efficiency (FE) are investigated by wide group of research community using several methods. From improving vehicle design and fleet management to supporting environmental initiatives and consumer applications, predictive models can drive innovation and efficiency across multiple sectors. By leveraging the power of deep learning (DL), organizations can make data-driven decisions that lead to significant economic, environmental and societal benefits. Predicting FE using Tensor Flow in Python with DL presents a powerful and efficient approach to addressing the challenges associated with fuel consumption and vehicle emissions. The combination of Tensor Flow's robust framework and the capabilities of DL allows for the development of highly accurate predictive models that can significantly impact various sectors. The continuous advancements in DL and the expanding capabilities of Tensor Flow promise further improvements in predictive accuracy and efficiency. As more data becomes available and computational power increases, the potential for even more sophisticated and impact applications of FE prediction will grow. Using Tensor Flow in Python with DL for predicting FE represents a significant step forward in the quest for more sustainable and efficient transportation solutions. By harnessing the power of advanced machine learning techniques, organizations can make data-driven decisions that lead to tangible benefits for the economy, the environment and society as a whole. This methodology not only enhances our ability to predict FE accurately but also opens up new avenues for innovation and improvement in various sectors, driving progress towards a more efficient and sustainable future.

KEYWORDS:

Fuel efficiency prediction; Machine Learning; Fuel consumption models; Deep learning

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1. Introduction

Driven by the need to optimize fuel consumption and reduce environmental impact across multiple industries, machine learning (ML) has become a major study field in fuel efficiency (FE) prediction. This paper performs a thorough review of fuel consumption prediction models with an emphasis on ML method implementation and comparison. The main goal is to investigate and determine the best ML strategies for fuel consumption prediction in various applications, such as commercial buildings, maritime vessels and fleet cars. Numerous approaches are used in the inquiry, such as statistical analysis, deep learning (DL) and supervised and unsupervised learning. The study assesses the accuracy, computational efficiency and generalizability of these models by utilizing extensive datasets gathered from real-world settings. To improve the models' predictive power, important fuel-consuming variables like driver behaviour, vehicle attributes and operating

circumstances are carefully examined [1]. The results demonstrate the advantage of some ML algorithms under particular conditions. Because DL techniques may identify intricate non-linear relationships in the data, for example, impressive accuracy when estimating fuel usage for maritime boats operating in a variety of settings. On the other hand, in situations where the data features are relatively simple, conventional ML methods like regression analysis and decision trees (DT) work well in forecasting fuel usage for fleet cars and commercial buildings [2].

The practical ramifications of applying ML-based fuel consumption prediction models are also covered in this study. It emphasizes how crucial it is to customize the models to particular use cases, taking into account elements like the availability of data, computing power and the solutions scalability. The study also promotes the use of hybrid techniques, which integrate domain-specific expertise and ML to maximize the benefits of both data-driven and expert-driven methodologies. The